

Next Feeding Session Predictor — Design Summary

Document: next-session-predictor-design-v2

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Core Energy Model

- **Unit:** Prepared milk ml (water ml × conversion factor, e.g. 90 ml water → 100 ml milk)
- **Daily target:** $\text{weightKg} \times 150 \text{ ml/kg/day}$
- **Hourly drain rate:** $\text{dailyTarget} / 24$
- **Milk balance:** tracks cumulative intake minus continuous drain — target is balance = 0

Water → Prepared Milk Conversion

Water ml	Prepared milk ml
30	35
60	70
90	100
120	135
150	170
180	200
210	240

Smoothed vs Strict 24h Tracking

- **Strict:** sums only feeds within the last 24h — drops abruptly when a feed crosses the boundary
- **Smoothed:** feeds past 24h retain decaying partial credit → much smoother curve; used for all calculations

Smoothed credit formula:

```
bottleCredit(age, milkMl) =  
    milkMl                                if age ≤ 24h  
    max(0, milkMl - hourlyRate × (age - 24))  if age > 24h
```

```
smoothed(T) =  $\sum$  bottleCredit(age_i, milkMl_i) for all feeds before T
```

Three Predictors

All predictors answer: **when should the next bottle be given?**

#	Name	Logic
P1	Standard	<code>lastFeed.start + milkMl / hourlyRate</code> — pure drain time
P2	Adjusted	Corrects P1 by $\pm 25\%$ based on current surplus/deficit
P3	Optimised (T*)	Binary search for the exact time where <code>smoothed(T*) + nextBottleMl = dailyTarget</code> — guarantees zero surplus

Default: P3 is on by default (`useTargetAwarePredictor = true`)

Predictor 1 — Standard

When has the energy from the current session been fully consumed?

```
standardNext = session.start + totalMilkMl / hourlyRate
```

Predictor 2 — Adjusted

Adjusts P1 to compensate for over/underfeeding:

```
surplus = smoothed - dailyTargetMl
rawCorrection = (surplus / hourlyRate) * 3600000 // ms
maxCorrectionMs = standardIntervalMs * (maxCorrectionPct / 100)

adjustedNext = standardNext + clamp(rawCorrection,
    -maxCorrectionMs, +maxCorrectionMs)
```

Predictor 3 — Optimised (T*)

Searches for the time T* where the existing pool has decayed to exactly `dailyTarget - nextBottleMl`, so giving one bottle brings the total to exactly the daily target:

```
targetBefore = dailyTargetMl - milkPerBottle
```

Binary search in $[lastFeed, lastFeed + maxCorrectionMs]$:

1. If $smoothed(lastFeed) \leq targetBefore \rightarrow T^* = lastFeed$ (feed now)
2. If $smoothed(T_{max}) > targetBefore \rightarrow T^* = T_{max}$ (cap applies)
3. Otherwise $\rightarrow$ **binary search** until $smoothed(T^*) = targetBefore$

App Design

Logging

- One entry per session: **start time** + total milk given
- Bottle size pre-fills the milk amount (editable if baby didn't finish)
- Always log the **start** of the session (when the first step was given)

Display Principle

The app always shows the energy state **at the moment of the last feed**, not at the current clock time:

```
smoothedAt = lastFeed.timestamp
```

- Status, bottle counts, and next feed times only update when a new feed is logged
- Only relative time labels ("in 2h 15m") update on clock ticks

Dashboard Layout

1. **Log Feed button**
2. **Daily Target card** (swipeable) — ml, bottle equivalent, interval
3. **Status card** (swipeable) — smoothed 24h intake, strict 24h intake, gauge, mood
4. **Three-column row** — Last feed | Next (standard) | Adjusted (T^*)
5. **Summary row** — total feeds, last 24h count, ml/hour

Settings

Setting	Default	Description
<code>weightKg</code>	—	Baby weight in kg

Setting	Default	Description
mlPerKgPerDay	150	Daily target formula (ml/kg/day)
displayBottleVolumeWater	90	Bottle size (water ml) for dashboard display
yellowThresholdPct	5	On-track zone: within $\pm 5\%$ of target
redThresholdPct	10	Seriously off: beyond $\pm 10\%$
timeFormat	24h	Time display format
maxCorrectionPct	25	Cap: max correction = $\pm 25\%$ of standard interval
useTargetAwarePredictor	true	true = P3 (T*), false = P2

Key Design Decisions

1. **Prepared milk ml** (not water ml) used for all energy calculations
2. **150 ml/kg/day** is the default daily target, configurable via settings
3. **Smoothed calculation** preferred over strict 24h sum to avoid abrupt fluctuations
4. **Predictor 3 (T*)** is the default — accounts for ongoing decay, guarantees zero surplus after feeding (within cap)
5. **Single log per session** (Option B) chosen over step-by-step logging for simplicity
6. **Display anchored to last feed time**, not current clock — handles delayed logging correctly
7. **$\pm 25\%$ correction cap** prevents extreme timing adjustments, spreading corrections across sessions